Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Oslo Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
20334 W Oslo RD	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/01/1971

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0045-0140-0028-0000

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: (209) 656-8434 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Nyman, Brad

Legal owner name: Nyman, Brad

Telephone no.: (209) 656-8434 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: -810,000 *gallons*

Total nitrogen generated: -3,744.74 *lbs*

Total phosphorus generated: -328.51 *lbs*

Total potassium generated: -7,367.80 *lbs*

Total salt generated: -29,336.01 *lbs*

	8,212,500	*gallons applied*
+	0	*gallons exported*
-	9,022,500	*gallons imported*
=	-810,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 6	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
01/01/2023	Process wastewater *WW Imported from Nyman Bros Dairy 10/19/22*	1,440,000.00 *gal*	543.00	60.70	907.00		7,300
02/26/2023	Process wastewater *WW Imported from Nyman Bros Dairy*	1,755,000.00 *gal*	714.00	101.00	640.00		4,260
05/16/2023	Process wastewater *WW Imported from Nyman Bros Dairy*	1,485,000.00 *gal*	554.00	48.60	1,090.00		4,340
07/13/2023	Process wastewater *WW Imported from Nyman Bros Dairy*	3,240,000.00 *gal*	554.00	48.60	1,090.00		4,340
08/11/2023	Process wastewater *WW Imported from Nyman Bros Dairy*	1,102,500.00 *gal*	168.00	60.10	713.00		3,900

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	40,371.94	4,677.86	69,811.04	357,120.52
Total imports for all materials	40,371.94	4,677.86	69,811.04	357,120.52

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	38	38	2	process wastewater	X045-X140-X022-XXXX
Field 2	6	6	2	process wastewater	X045-X140-X028-XXXX
Field 3	7	7	2	process wastewater	X045-X140-X028-XXXX
Field 4	35	35	2	process wastewater	X045-X140-X035-XXXX X045-X140-X036-XXXX
Totals for areas that were used for application	86	86	8		
Totals for areas that were not used for application					
Land application area totals	86	86	8		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	570.00 *ton*	As-is		70.0	6,450.00	1,470.00	16,800.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.00	193.50	44.10	504.00	990.00

Field 1

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,015.00 *ton*	As-is		69.0	3,290.00	1,000.00	3,130.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	26.71	175.76	53.42	167.21	844.59

Field 2

Field name: Field 2

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	90.00 *ton*	As-is		70.0	6,840.00	1,380.00	13,800.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.00	205.20	41.40	414.00	864.00

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	150.00 *ton*	As-is		64.2	4,050.00	1,100.00	3,520.00		3.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.00	202.50	55.00	176.00	698.10

Field 3

Field name: Field 3

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	115.00 *ton*	As-is		70.0	5,760.00	1,410.00	16,500.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.43	189.26	46.33	542.14	985.71

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	180.00 *ton*	As-is		65.0	3,780.00	1,070.00	3,860.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.71	194.40	55.03	198.51	738.00

Field 4

Field name: Field 4

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	510.00 *ton*	As-is		70.0	6,180.00	1,440.00	19,800.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.57	180.10	41.97	577.03	1,136.57

Field 4

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	895.00 *ton*	As-is		66.6	3,770.00	1,070.00	3,860.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.57	192.81	54.72	197.41	700.35

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/05/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.76	7.80	116.52	937.82	585,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	248.59	5,265,000.00 *gal*
Application event totals		69.76	7.80	116.52	1,186.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.89	15.97	101.19	673.57	720,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	305.95	6,480,000.00 *gal*
Application event totals		112.89	15.97	101.19	979.53	

Field 1 - 05/25/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.65	6.72	150.80	600.44	630,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	267.71	5,670,000.00 *gal*
Application event totals		76.65	6.72	150.80	868.15	

Field 1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	267.71	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	267.71	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	267.71	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	267.71	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.70	5.76	129.26	514.67	540,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		65.70	5.76	129.26	744.13	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.75	4.80	107.72	428.89	450,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	191.22	4,050,000.00 *gal*
Application event totals		54.75	4.80	107.72	620.11	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	229.47	

Field 1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.60	5.94	70.46	385.41	450,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	191.22	4,050,000.00 *gal*
Application event totals		16.60	5.94	70.46	576.63	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	152.98	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	152.98	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	191.22	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	191.22	

Field 2 - 11/05/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.97	7.60	113.53	913.78	90,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		67.97	7.60	113.53	1,155.99	

Field 2 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.06	18.96	120.17	799.87	135,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	363.32	1,215,000.00 *gal*
Application event totals		134.06	18.96	120.17	1,163.19	

Field 2 - 05/25/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.02	9.13	204.66	814.89	135,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	363.32	1,215,000.00 *gal*
Application event totals		104.02	9.13	204.66	1,178.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	302.77	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	302.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	242.21	

Field 2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	90,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		69.35	6.08	136.44	785.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	90,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		69.35	6.08	136.44	785.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	242.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.03	7.52	89.25	488.18	90,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		21.03	7.52	89.25	730.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	181.66	607,500.00 *gal*
Application event totals		0.00	0.00	0.00	181.66	

Field 2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	242.21	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	242.21	

Field 3 - 11/05/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.39	9.77	145.97	1,174.86	135,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		87.39	9.77	145.97	1,434.37	

02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	153.21	21.67	137.33	914.14	180,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	415.22	1,620,000.00 *gal*
Application event totals		153.21	21.67	137.33	1,329.36	

Field 3 - 05/25/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.16	7.82	175.42	698.48	135,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	311.42	1,215,000.00 *gal*
Application event totals		89.16	7.82	175.42	1,009.89	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.51	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.51	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.30	6.52	146.19	582.06	112,500.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		74.30	6.52	146.19	841.58	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.30	6.52	146.19	582.06	112,500.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		74.30	6.52	146.19	841.58	

Field 3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.53	8.06	95.62	523.05	112,500.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		22.53	8.06	95.62	782.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	259.51	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.51	

Field 4 - 11/05/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.56	9.12	136.24	1,096.53	630,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	5,670,000.00 *gal*
Application event totals		81.56	9.12	136.24	1,387.19	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.57	17.34	109.87	731.31	720,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	332.18	6,480,000.00 *gal*
Application event totals		122.57	17.34	109.87	1,063.49	

Field 4 - 05/25/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.27	6.78	152.03	605.35	585,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	269.90	5,265,000.00 *gal*
Application event totals		77.27	6.78	152.03	875.24	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Field 4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	249.13	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	249.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.33	6.26	140.34	558.78	540,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	249.13	4,860,000.00 *gal*
Application event totals		71.33	6.26	140.34	807.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.38	5.74	128.64	512.22	495,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	228.37	4,455,000.00 *gal*
Application event totals		65.38	5.74	128.64	740.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	249.13	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	249.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.03	6.45	76.50	418.44	450,000.00 *gal*
TID Lateral 6	Surface water	0.00	0.00	0.00	207.61	4,050,000.00 *gal*
Application event totals		18.03	6.45	76.50	626.05	

Field 4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	207.61	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.61	

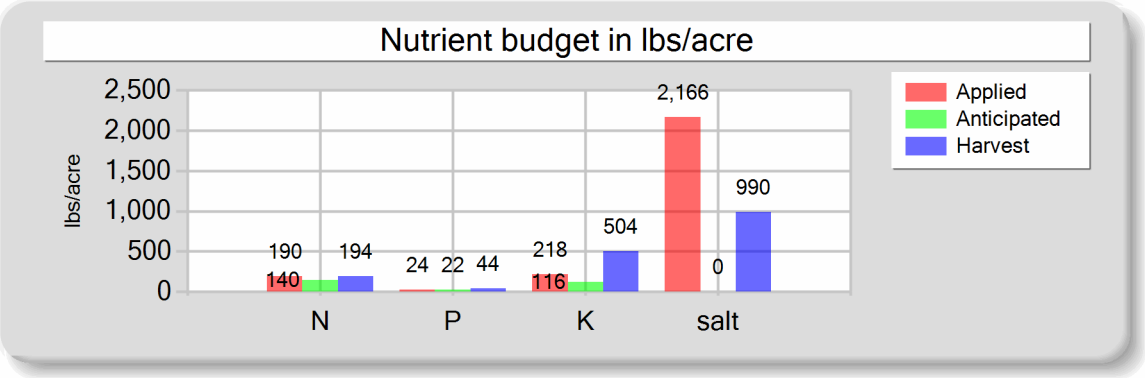
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	207.61	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.61	

B. NUTRIENT BUDGET

Field 1 - 11/05/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.65	23.77	217.72	1,611.40
Fresh water	0.00	0.00	0.00	554.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.65	23.77	217.72	2,165.94
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	193.50	44.10	504.00	990.00
Nutrient balance	-3.85	-20.33	-286.28	1,175.94
Applied to removed ratio	0.98	0.54	0.43	2.19

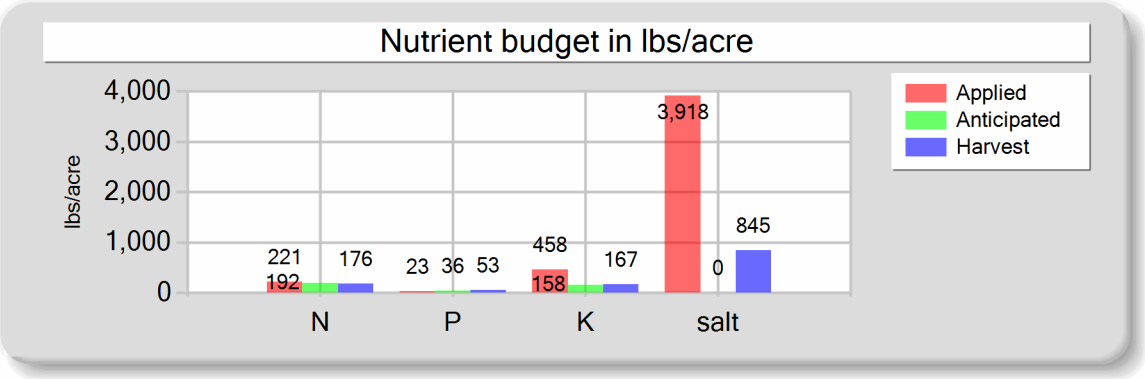
Fresh water applied	
11,745,000.00	*gallons*
432.53	*acre-inches*
11.38	*inches/acre*

Process wastewater applied	
1,305,000.00	*gallons*
48.06	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/25/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.69	23.23	458.24	1,929.41
Fresh water	0.00	0.00	0.00	1,988.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.69	23.23	458.24	3,918.11
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	175.76	53.42	167.21	844.59
Nutrient balance	44.94	-30.19	291.03	3,073.52
Applied to removed ratio	1.26	0.43	2.74	4.64

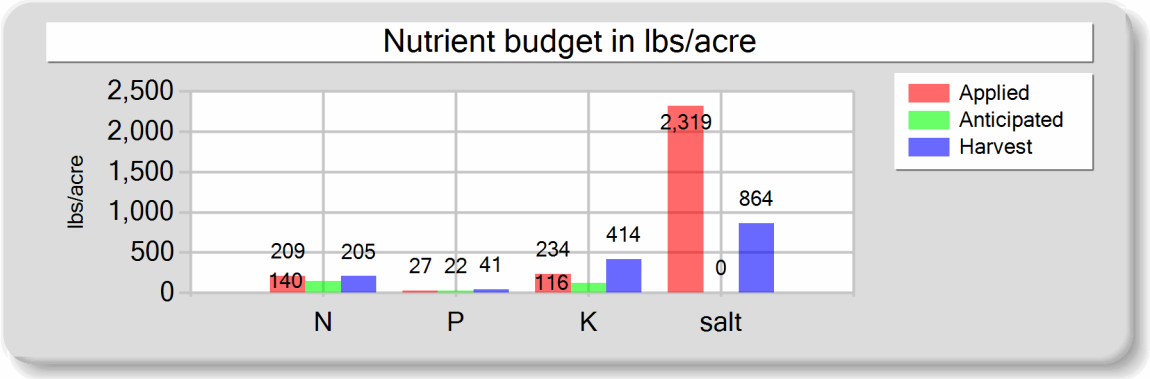
Fresh water applied	
42,120,000.00	*gallons*
1,551.14	*acre-inches*
40.82	*inches/acre*

Process wastewater applied	
2,070,000.00	*gallons*
76.23	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/05/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.03	26.56	233.70	1,713.65
Fresh water	0.00	0.00	0.00	605.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.03	26.56	233.70	2,319.18
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	205.20	41.40	414.00	864.00
Nutrient balance	3.83	-14.84	-180.30	1,455.18
Applied to removed ratio	1.02	0.64	0.56	2.68

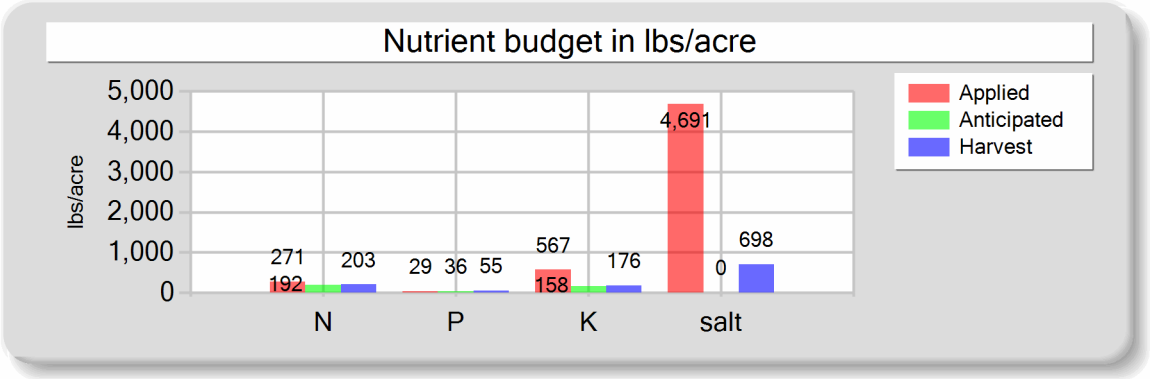
Fresh water applied	
2,025,000.00	*gallons*
74.57	*acre-inches*
12.43	*inches/acre*

Process wastewater applied	
225,000.00	*gallons*
8.29	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/25/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.74	28.82	566.79	2,389.59
Fresh water	0.00	0.00	0.00	2,301.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.74	28.82	566.79	4,690.62
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	202.50	55.00	176.00	698.10
Nutrient balance	68.24	-26.18	390.79	3,992.52
Applied to removed ratio	1.34	0.52	3.22	6.72

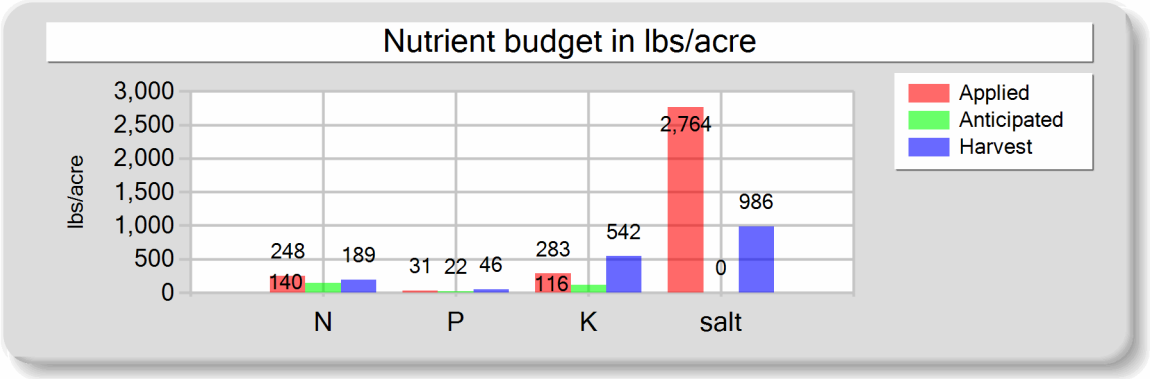
Fresh water applied	
7,695,000.00	*gallons*
283.38	*acre-inches*
47.23	*inches/acre*

Process wastewater applied	
405,000.00	*gallons*
14.91	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/05/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.60	31.44	283.31	2,088.99
Fresh water	0.00	0.00	0.00	674.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.60	31.44	283.31	2,763.73
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	189.26	46.33	542.14	985.71
Nutrient balance	58.35	-14.89	-258.84	1,778.02
Applied to removed ratio	1.31	0.68	0.52	2.80

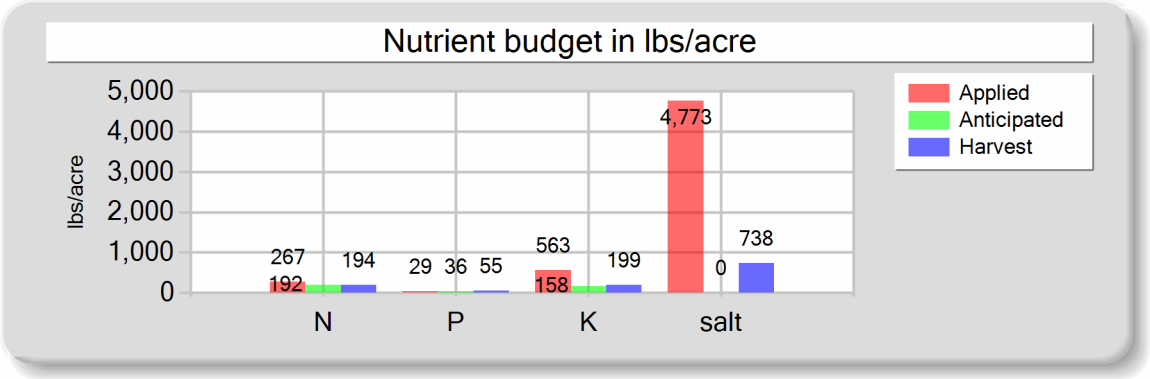
Fresh water applied	
2,632,500.00	*gallons*
96.95	*acre-inches*
13.85	*inches/acre*

Process wastewater applied	
315,000.00	*gallons*
11.60	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/25/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	260.29	28.92	563.42	2,385.66
Fresh water	0.00	0.00	0.00	2,387.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.29	28.92	563.42	4,773.19
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	194.40	55.03	198.51	738.00
Nutrient balance	72.89	-26.11	364.91	4,035.19
Applied to removed ratio	1.37	0.53	2.84	6.47

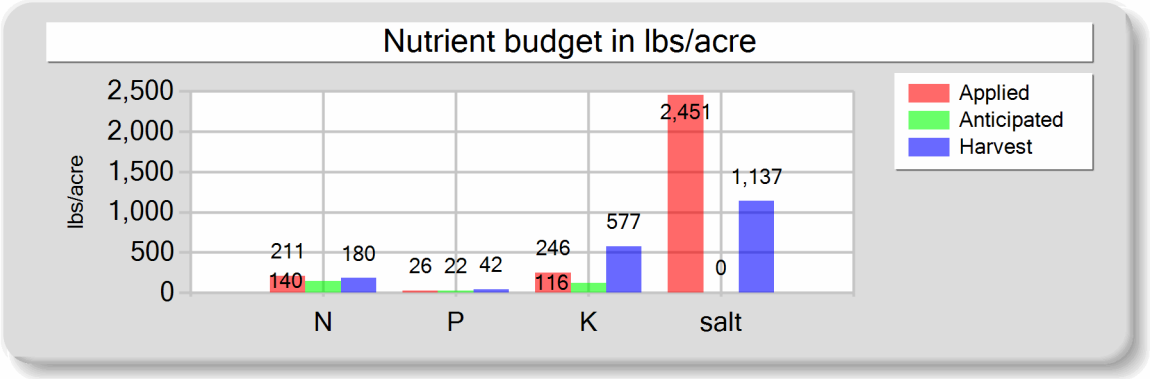
Fresh water applied	
9,315,000.00	*gallons*
343.04	*acre-inches*
49.01	*inches/acre*

Process wastewater applied	
472,500.00	*gallons*
17.40	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/05/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.14	26.46	246.11	1,827.84
Fresh water	0.00	0.00	0.00	622.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.14	26.46	246.11	2,450.68
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	180.10	41.97	577.03	1,136.57
Nutrient balance	31.03	-15.51	-330.92	1,314.10
Applied to removed ratio	1.17	0.63	0.43	2.16

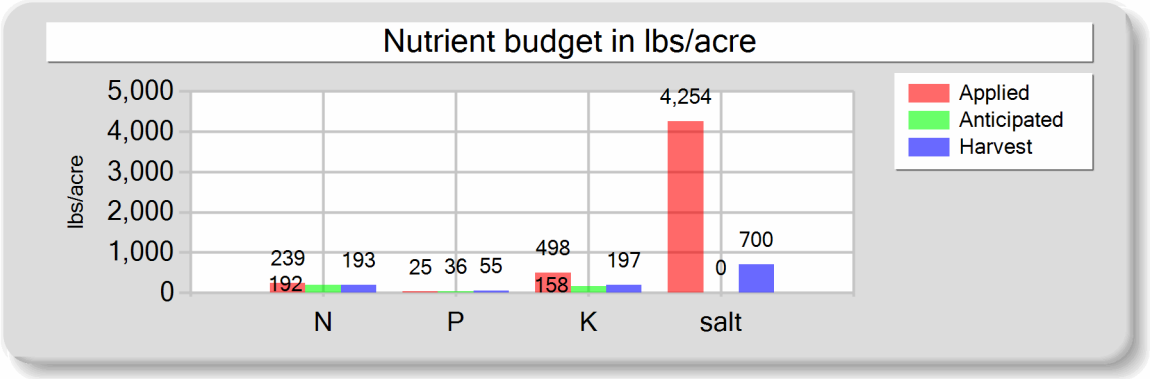
Fresh water applied	
12,150,000.00	*gallons*
447.44	*acre-inches*
12.78	*inches/acre*

Process wastewater applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
1.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/25/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.01	25.22	497.52	2,094.79
Fresh water	0.00	0.00	0.00	2,159.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.01	25.22	497.52	4,253.95
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	192.81	54.72	197.41	700.35
Nutrient balance	46.20	-29.50	300.11	3,553.60
Applied to removed ratio	1.24	0.46	2.52	6.07

Fresh water applied	
42,120,000.00	*gallons*
1,551.14	*acre-inches*
44.32	*inches/acre*

Process wastewater applied	
2,070,000.00	*gallons*
76.23	*acre-inches*
2.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 17.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	2,030.00	3,550.00							49.40
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 10/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 48.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	3,730.00	6,480.00	10,600.00	3,510.00	2,110.00	3,850.00	729.00		44.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4
Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	543.00	358.00	0.00	0.00	60.70	907.00								7,180.00	7,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	714.00	339.00	0.00	0.00	101.00	640.00								5,560.00	4,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	384.00	0.00	0.00	48.60	1,090.00								7,740.00	4,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	168.00	104.00	0.00	0.00	60.10	713.00	86.90	81.50	151.00	1,700.00	0.00	139.00	322.00	4,200.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	22.40	5.60	0.00	0.00	3.80	81.70	166.00	57.90	106.00	500.00	0.00	205.00	118.00	1,970.00	1,140
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	434.00	291.00	0.00	0.00	33.60	491.00								5,110.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	426.00	308.00	0.00	0.00	34.90	490.00								5,170.00	4,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 6

TID Lateral 6

Sample description: TID Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,450.00	1,470.00	16,800.00		11.00
DL	100.00	100.00	100.00		0.01

Field 1 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	1,000.00	3,130.00		5.10
DL	100.00	100.00	100.00		0.01

Field 2 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,840.00	1,380.00	13,800.00		9.60
DL	100.00	100.00	100.00		0.01

Field 2 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,050.00	1,100.00	3,520.00		3.90
DL	100.00	100.00	100.00		0.01

Field 3 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,760.00	1,410.00	16,500.00		10.00
DL	100.00	100.00	100.00		0.01

Field 3 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,780.00	1,070.00	3,860.00		4.10
DL	100.00	100.00	100.00		0.01

Field 4 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,180.00	1,440.00	19,800.00		13.00
DL	100.00	100.00	100.00		0.01

Field 4 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,770.00	1,070.00	3,860.00		4.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

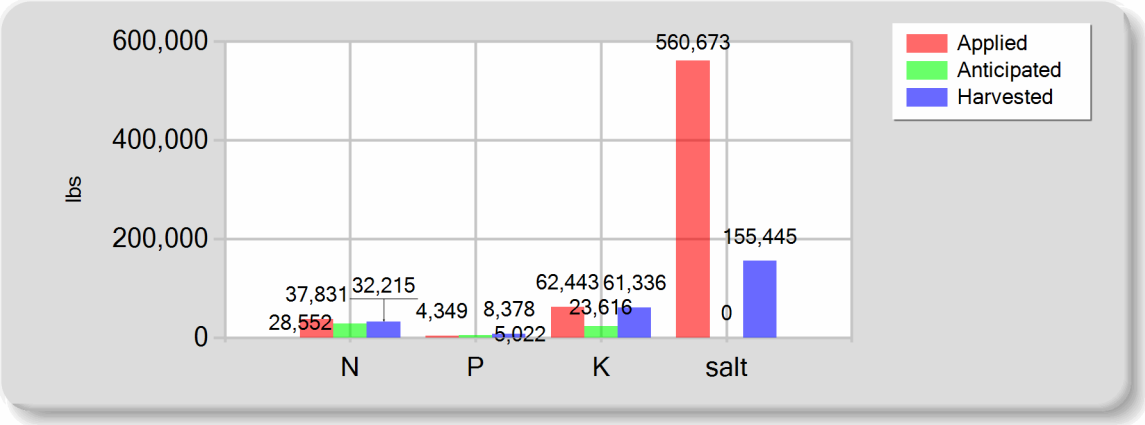
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

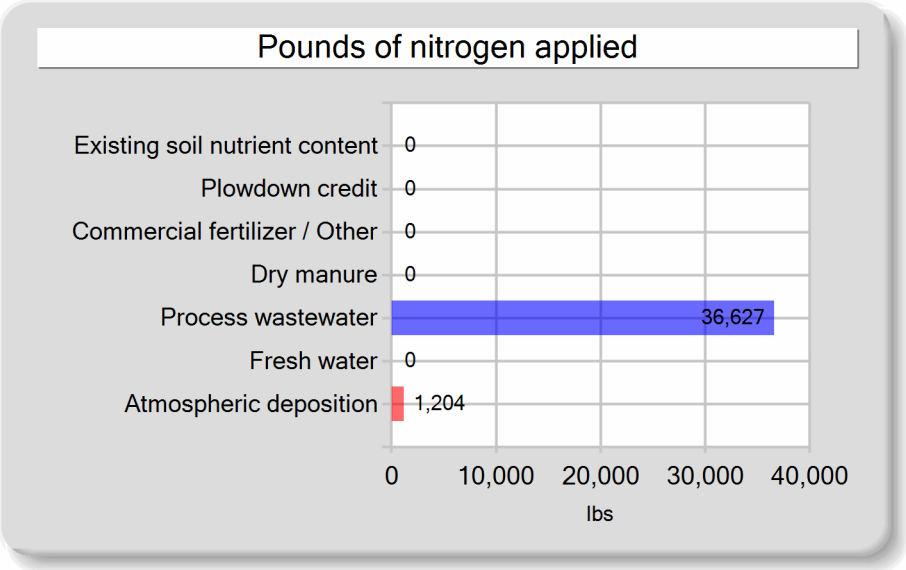
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

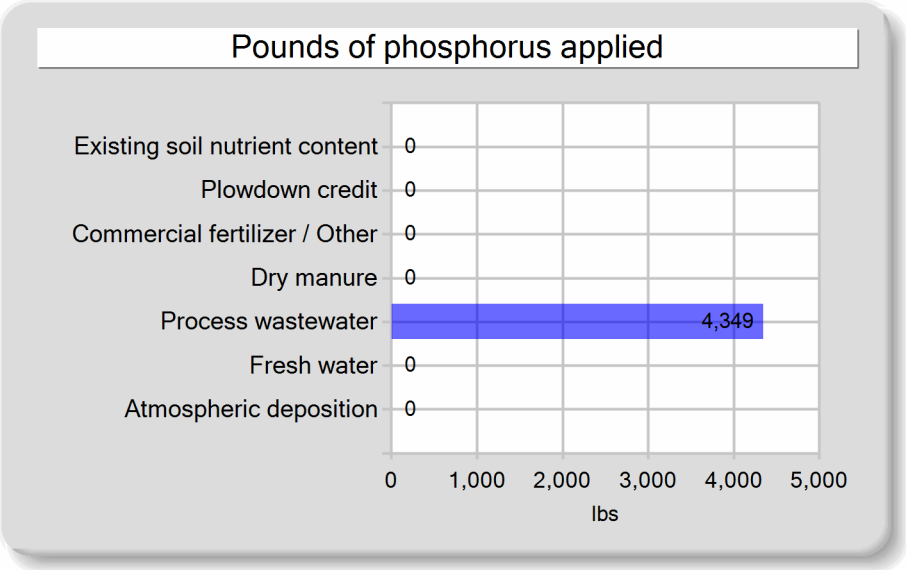
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	36,627.21	4,349.35	62,443.24	327,784.51
Fresh water	0.00	0.00	0.00	232,888.40
Atmospheric deposition	1,204.00	0.00	0.00	0.00
Total nutrients applied	37,831.21	4,349.35	62,443.24	560,672.91
Anticipated crop nutrient removal	28,552.00	5,022.40	23,615.60	0.00
Actual crop nutrient removal	32,215.40	8,377.80	61,335.90	155,445.16
Nutrient balance	5,615.81	-4,028.45	1,107.34	405,227.75
Applied to removed ratio	1.17	0.52	1.02	3.61

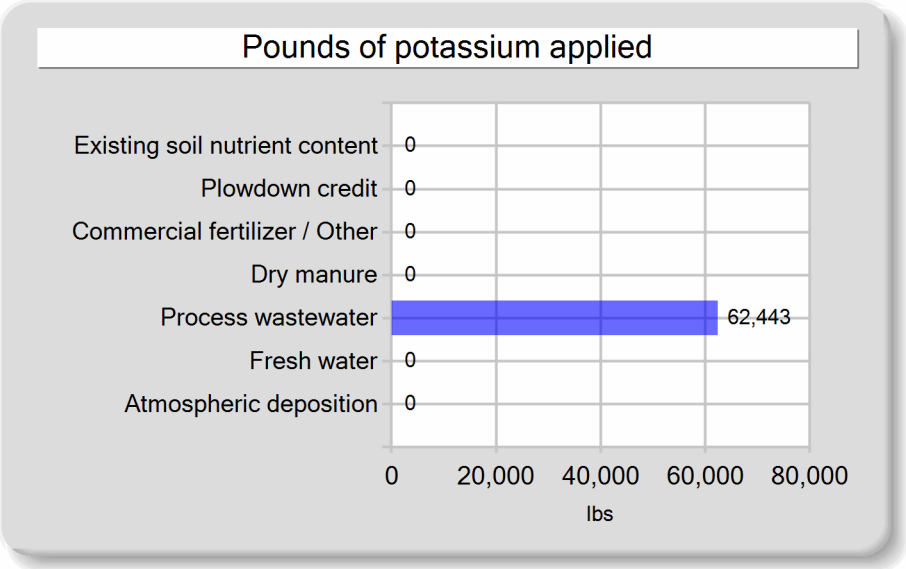
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

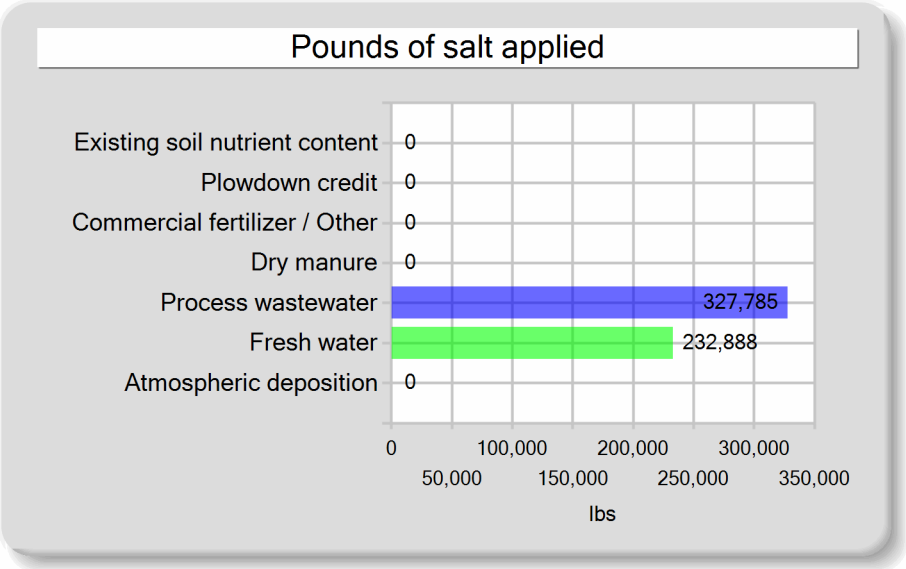


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:31 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nyman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
OSLO DAIRY
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3240032A	Electrical Conductivity (EC)	EPA 120.1 and	2.03	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1300	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
OSLO DAIRY
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3240032B	Electrical Conductivity (EC)	EPA 120.1 and	1.16	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	34.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	744	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.1	102 (T3240032B)	89.6 (T3240032B)	12.6

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Oslo
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To:
FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/24	1:00	Domestic	DW1	15.6°		✓				
2	S	1:15	wells	DW2	16.0°		✓				
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FRAC				
2nd		DENELE ANALYTICAL, INC.	8-28-23 12:36			T3240032

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Oslo 2023 Annual Report

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAcw1sL1-M-VG38ZY3uvVFITXmX9J0qJbV

"Oslo 2023 Annual Report" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-06-19 - 4:04:29 PM GMT

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